

4(a)	any five from: 1 a gene codes for a protein which determines the phenotype ; 2 <u>F8</u> gene ; 3 located on X chromosome / sex-linked ; 4 <i>ref. to</i> recessive allele ; 5 non-functioning factor VIII (protein) or less factor VIII (protein) or no factor VIII (protein) ; 6 (so) blood does not clot quickly enough / excessive bleeding occurs (after an injury) ; 7 link the phenotype to the genotype ; e.g. $X^F X^F$ = normal female $X^F X^f$ = normal female $X^f X^f$ = affected female $X^F Y$ = normal male $X^f Y$ = affected male 8 AVP ; e.g. prevents activation of thrombin / fibrinogen not converted to fibrin / <i>ref. to</i> mostly males affected / different mutations cause a range of severity of haemophilia	5
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Question	Answer	Marks
4(b)	any three from: 1 <i>lele</i> is, homozygous recessive / two recessive alleles ; 2 <i>le</i> (allele) / <i>lele</i>, codes for a non-functional <u>enzyme</u> ; 3 no, active gibberellin / GA₁ or <i>ref. to</i> gibberellin inactive ; 4 no / less, cell elongation or no / less, stem elongation / increase in internodal length ; 5 AVP ; e.g. <i>ref. to</i> GA 3β-hydroxylase / GA3-oxidase / last enzyme in the pathway that produces, active gibberellin / GA₁ <i>le</i> allele codes for alanine instead of threonine (in the enzyme)	3
4(c)	any three from: 1 (DELLA proteins) bind to, PIF / transcription factor ; 2 PIF / transcription factor, cannot bind to, DNA / promoter ; 3 RNA polymerase cannot bind to <u>promoter</u> ; 4 no transcription / mRNA not made ;	3